

Polyomino rectangles by parity: no rectangle for 8, 10 or 16 cells

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Abstract

For which n can one copy of every hole-free n -omino tile a rectangle? On MathOverflow the answer is known to be yes for $n = 1, 2, 5, 7$, no for $n = 3, 4, 6$ and no for $n \geq 17$; the cases $8 \leq n \leq 16$ were open. We show that the answer is no for $n = 8, 10$ and 16 . The reason is the checkerboard count that already rules out 4 and 6: with n even, the imbalances of the pieces must add up to a multiple of 4, and for these three sizes their total is 2 modulo 4. The totals come from two independent enumerations of the 363, 4460 and 11 230 003 pieces. For $n = 12, 14$ and every odd n the count gives no obstruction, so 9, 11, 12, 13, 14 and 15 remain open. The argument is checked in the Lean proof assistant.

1 The question

An n -omino is a connected set of n unit squares of the grid, and it is hole-free if its complement is connected. Ralph Morrison asked on MathOverflow [1] for which n one copy of each hole-free n -omino, up to rotation and reflection, can tile a rectangle. The answer is yes for $n = 1, 2, 5$ and 7 , by explicit tilings, and no for $n = 3, 4$ and 6 . The answers to the question show that it is no for every $n \geq 18$ (Budd and Lehner) and, by a computer count, for $n = 17$ (Hoffman). That left $8 \leq n \leq 16$.

Here is the case $n = 4$. Colour the grid like a chessboard. Four of the five tetrominoes cover two black and two white squares wherever they are put; the T-tetromino covers three of one colour and one of the other (Figure 1). Stop and decide before reading on: why can the five never fill a rectangle?

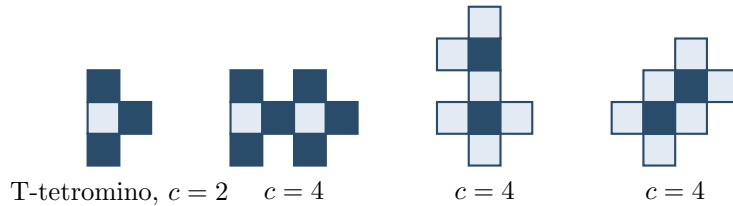


Figure 1: The T-tetromino, and the three octominoes that are furthest from balanced, coloured like a chessboard. The imbalance c is the number of dark squares minus the number of light ones, up to sign; for the T-tetromino it is 2, for these octominoes 4, the largest of any octomino.

A rectangle of area 20 has an even side, so it has as many black squares as white ones. The four balanced pieces leave the balance unchanged, and the T-tetromino shifts it by 2 one way or the other. So the five pieces can never cover a balanced region. This is the parity argument the question refers to, and Golomb used the same count to show that the 35 hexominoes tile no rectangle (see [2, Example 3]). Our result is that the same count, applied to all the pieces at once, also settles three of the open sizes.

Theorem 1. *For $n = 8, 10$ and 16 , the hole-free n -ominoes, one of each, tile no rectangle.*

2 The checkerboard count

Give the square (x, y) the colour $\chi(x, y) = (-1)^{x+y}$. The *imbalance* of a finite set of squares is the sum of χ over it, and for a piece P we write $c(P)$ for its absolute value. A placed piece is the image of P under a rotation or reflection of the grid followed by a translation.

Lemma 1. *A placed copy of P has imbalance $c(P)$ or $-c(P)$, and $c(P)$ has the same parity as the number of squares of P .*

Proof. A rotation or reflection of the grid preserves the parity of $x + y$, so it preserves χ . A translation by (a, b) multiplies χ by $(-1)^{a+b}$. For the parity: each square contributes ± 1 , which is odd. $\square$

Lemma 2. *A rectangle with an even side has imbalance 0.*

Proof. Two neighbouring rows, or two neighbouring columns, have opposite colours square by square. $\square$

Proposition 1. *Let $P_1, \dots, P_N$ be pieces with an even number of squares each, and suppose*

$$c(P_1) + c(P_2) + \dots + c(P_N) \equiv 2 \pmod{4}.$$

Then no rectangle is tiled by placed copies of $P_1, \dots, P_N$, one of each.

Proof. The area is even, so the rectangle has an even side and imbalance 0 by Lemma 2. Adding up over the pieces, Lemma 1 gives signs $\varepsilon_i = \pm 1$ with

$$\varepsilon_1 c(P_1) + \varepsilon_2 c(P_2) + \dots + \varepsilon_N c(P_N) = 0.$$

Each $c(P_i)$ is even, by Lemma 1, and for an even number c we have $-c \equiv c \pmod{4}$. So the left-hand side is congruent to $\sum c(P_i) \equiv 2 \pmod{4}$, and cannot be 0. $\square$

The proposition says nothing about which rectangle, and nothing about the shapes beyond their imbalances. All that is left is to add up $c(P)$ over the hole-free n -ominoes, which we write $S(n)$.

3 The totals

Table 1 gives $S(n)$ for $4 \leq n \leq 16$. For even n it is 2 modulo 4 exactly when $n \in \{4, 6, 8, 10, 16\}$, which proves Theorem 1 by Proposition 1.

The totals come from enumerating pieces. The first program grows the fixed n -ominoes (those counted without rotating or reflecting) by Redelmeier's method [3], tests each for holes and computes its imbalance and the number s of the eight symmetries that fix it. A free piece is an orbit of $8/s$ fixed ones, so $S(n)$ is the sum of sc over hole-free fixed pieces, divided by 8. The second program shares no code with the first: it grows the free pieces themselves, one square at a time, keeping one representative of each in a hash table. Both reproduce the counts in OEIS A001168, A000105 and A000104 for every $n \leq 16$, and they agree on every $S(n)$ in the table. For $n \leq 10$ a third, pure Python enumeration agrees as well.

Where the count is silent. For odd n every $c(P)$ is odd and $-c \not\equiv c \pmod{4}$, so the congruence gives nothing. For $n = 12$ and 14 the total is 0 modulo 4. In all these cases we checked that some choice of signs does reach the imbalance the rectangle needs (0, or ± 1 when the area is odd), so the checkerboard alone cannot decide them.

n	hole-free n -ominoes	$S(n)$	$S(n) \bmod 4$	checkerboard
4	5	2	2	rules out (known)
5	12	14	2	no obstruction (n odd)
6	35	22	2	rules out (known)
7	107	123	3	no obstruction (n odd)
8	363	278	2	rules out
9	1 248	1 502	2	no obstruction (n odd)
10	4 460	4 010	2	rules out
11	16 094	20 462	2	no obstruction (n odd)
12	58 937	59 248	0	no obstruction
13	217 117	291 211	3	no obstruction (n odd)
14	805 475	886 436	0	no obstruction
15	3 001 127	4 231 527	3	no obstruction (n odd)
16	11 230 003	13 329 078	2	rules out

Table 1: Hole-free n -ominoes (OEIS A000104 [4]) and the sum $S(n)$ of their checkerboard imbalances.

4 Where the question stands

Combining Theorem 1 with the earlier results, one copy of each hole-free n -omino tiles a rectangle for $n = 1, 2, 5, 7$ and for no other n outside $\{9, 11, 12, 13, 14, 15\}$. Whether the cavity argument that settles $n \geq 17$ reaches any of them is not known, and a construction would have to arrange more than a thousand pieces. Other colourings, of the kind Garvie and Burkardt search for systematically [2], are a natural next test.

Verification. Lemmas 1 and 2 and Proposition 1 are proved in Lean 4 with Mathlib [5], using only the standard axioms, for pieces placed by any of the eight symmetries and any translation, with disjointness of the placed pieces as the only assumption on the tiling. The Lean statement of Theorem 1 takes the three totals from Table 1 as hypotheses; they are computed, not proved, by the two independent programs. Python scripts rerun both programs, compare the counts with OEIS, check the table and the sign choices for $n = 12, 14$ and odd n , and reject deliberately broken versions of each program. Everything is at <https://github.com/jvvk/mathematics/tree/main/polyomino-rectangle-parity>.

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References

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